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SUMMATIVE ASSESSMENT - I 2026-27

PHYSICS — STANDARD X

Section - A

1. Which of the following is a longitudinal wave?

a) Radio wave

b) Sound wave • (Correct Answer)

c) Wave on the surface of water

d) Light wave

Explanation: *Sound waves travel through medium particles as back-and-forth vibrations along the direction of propagation, making them longitudinal waves. The others are transverse waves.*

2. Assertion (A): The walls of large halls like cinema theatres are made rough.
Reason (R): Rough surfaces reflect sound more effectively than smooth surfaces.

a) A is correct, R is incorrect • (Correct Answer)

b) Both A and R are incorrect

c) A is incorrect, R is correct

d) Both A and R are correct

Explanation: *Rough surfaces cause diffuse reflection, absorbing or scattering sound evenly to minimize echo and reverberation. Smooth surfaces reflect sound more orderly and effectively, which creates undesirable echoes.*

3. Statements regarding image formation by a telescope are given below:
i) The image formed by the objective is always erect.
ii) The image formed by the objective is small and real.
iii) The position of the image formed by the objective is in between F and 2F of the eyepiece.
iv) The image formed by the eyepiece is virtual.

Which of the following is correct?

a) i & iii

b) ii & iv • (Correct Answer)

c) i & iv

d) iii & iv

Explanation: *In an astronomical telescope, the objective lens forms a real, inverted, and diminished (small) image of a distant object at its focal plane. The eyepiece then treats this image as an object within its focal length to produce a highly magnified virtual final image.*

Question 4

Some statements related to lenses are given. Match the items suitably in column 1 with column 2.

Column 1	Column 2

A. Light ray parallel to the optic axis falling on a concave lens	I. Passes parallel to the optic axis
B. Light ray passing through optic centre	II. Passes through the principal focus
C. Light ray parallel to the optic axis falling on a convex lens	III. Appears to come from the principal focus and moves away from the optic axis
D. Light ray passing through a concave lens directed towards the principal focus on the other side	IV. Passes along a straight line without undergoing refraction

Options Given in Paper:
a) A - IV, B - III, C - I, D - II b) A - III, B - IV, C - II, D - I
c) A - II, B - III, C - IV, D - I d) A - IV, B - III, C - II, D - I

Answer: (b) A - III, B - IV, C - II, D - I

Section - B

Question 5

A, B are two swings hung from an iron rod fixed as shown in the figure (Swing A length = 4m, Swing B length = 6m). Which swing will complete 10 oscillations faster, if they are allowed to oscillate? Justify your answer.

Answer: Swing A will complete 10 oscillations faster.

> ***Justification:*** *The time period (T) of a simple pendulum/swing depends on its length (l) given by the formula $T = 2\pi\sqrt{(l/g)}$. Since Swing A has a shorter length (4 m < 6 m), its time period is smaller, meaning it completes each oscillation in less time and vibrates faster.*

Question 6

If you are clapping in front of a hill, standing away from it, you can hear the sound again after a short time.
a) Name this phenomenon.
b) Can you hear the sound distinctly again if you clap standing close to the hill? Explain why?

Answer:
a) Echo.
b) No, you cannot hear the sound distinctly.

> ***Explanation:*** *To hear a distinct echo, the time interval between the original sound and the reflected sound must be at least 0.1 seconds (persistence of hearing). If you stand close to the hill, the distance is short, so the reflected sound returns in less than 0.1 seconds and merges with the original sound.*

Question 7

Low magnification of the image formed by the objective of a compound microscope is not beneficial.
a) Why?
b) Suggest a solution for this.

Answer:
a) The total magnification of a compound microscope is the product of the magnification of the objective lens and the eyepiece ($M = m_o \times m_e$). If the objective lens provides low magnification, the overall magnifying power of the microscope will remain insufficient.
b) Solution: Use an objective lens with a very short focal length (f_o), which inherently increases its magnifying power.

Question 8

An object on the principal axis of a convex lens is 10 cm away from optic centre. The height of the object is 3 cm. The focal length of the lens is 4 cm. Draw a ray diagram of image formation, based on the given measurements.

Answer Calculation:
Using the lens formula: $1/f = 1/v - 1/u$
Given: $f = +4$ cm, $u = -10$ cm, $h_o = 3$ cm
$1/v = 1/4 + 1/(-10) = (5 - 2)/20 = 3/20 \Rightarrow v = +20/3$ cm $\approx +6.67$ cm (Real image formed on the other side).
Magnification: $m = v/u = (20/3) / (-10) = -2/3$
Height of image: $h_i = m \times h_o = (-2/3) \times 3 = -2$ cm (Inverted image of height 2 cm).

Question 9

Observe the figure of a wave motion.
a) What is the amplitude of this wave?
b) Find the time required for the wave to travel 24 m if the speed of wave is 8 m/s.

Answer:
a) Amplitude = 0.2 m
b) Time = 3 seconds

Calculation for part (b): *Time = Distance / Speed = 24 m / 8 m/s = 3 s.*

Question 10. A.

An object of height 1 cm is placed 25 cm away from a convex lens with a focal length of 20 cm.
a) How far the image is formed from the lens.
b) Find the height of this image.

Answer / Calculation:
Given Data:
• Height of object (h_o) = +1 cm
• Object distance (u) = -25 cm
• Focal length (f) = +20 cm (convex lens)

a) Image Distance (v):
Using Lens Formula: $1/f = 1/v - 1/u$
$1/20 = 1/v - (1/-25) \Rightarrow 1/20 = 1/v + 1/25$
$1/v = 1/20 - 1/25 = (5 - 4) / 100 = 1/100 \Rightarrow v = +100$ cm
The image is formed at a distance of 100 cm on the opposite side of the lens.

b) Height of Image (h_i):
Using Magnification: $m = v/u = h_i / h_o$
$100 / -25 = h_i / 1 \Rightarrow -4 = h_i \Rightarrow h_i = -4$ cm
The height of the image is 4 cm (real and inverted).

Question 10. B. (OR)

An object of height 2 cm is placed in front of a lens. A real image of height 5 cm is obtained 10 cm away from the lens. Find the:
a) Magnification of this image.
b) Distance of the object from the lens.

Answer / Calculation:
Given Data:
• Height of object (h_o) = +2 cm

• Height of real image (h_i) = -5 cm (inverted)
• Image distance (v) = +10 cm (real image on other side)

a) Magnification (m):
$m = h_i / h_o = -5 / 2 = -2.5$

b) Object Distance (u):
$m = v/u \Rightarrow -2.5 = 10 / u \Rightarrow u = 10 / -2.5 = -4$ cm
The object is placed at a distance of 4 cm in front of the lens.

Section - C

Question 11. A.

Lower the focal length of a lens, greater its power. Do you agree with this statement? Justify your answer using simple mathematical operations.

Answer: Yes, I agree with this statement.

Justification: *Power (P) is inversely proportional to focal length (f) in meters: P = 1/f*
• Case 1: If f = 0.2 m, then P = 1 / 0.2 = 5 D
• Case 2: If f = 0.1 m (lower focal length), then P = 1 / 0.1 = 10 D
Since 10 D > 5 D, a lower focal length results in a greater power.

Question 11. B. (OR)

An image is formed 100 cm away from the optic centre of a lens with power +2 D. A child remarked that the size of the image is half the size of the object. Do you agree with this. Justify your answer.

Answer: No, I do not agree with the child's remark.

Justification:
• Focal length f = 1/P = 1/+2 = 0.5 m = +50 cm
• Given image distance v = +100 cm (assuming a real image since it's formed at 2f).
• Lens formula: $1/f = 1/v - 1/u \Rightarrow 1/50 = 1/100 - 1/u$
• $1/u = 1/100 - 1/50 = -1/100 \Rightarrow u = -100$ cm
• Magnification: $m = v/u = 100 / -100 = -1$
Since $|m| = 1$, the size of the image is equal to the size of the object, not half.

Question 12. A.

A tuning fork has a natural frequency 288 Hz. When this tuning fork is excited and its stem is pressed on a table of natural frequency 88 Hz, the table also vibrates.
a) What do you mean by natural frequency?
b) Why does the table vibrate?
c) What is the frequency at which the table vibrates?

Answers:
a) Natural Frequency: It is the constant frequency at which a body vibrates when it is disturbed and allowed to oscillate freely without any external periodic force.
b) Table Vibration: The table is forced to vibrate due to the periodic transmission of vibrations from the stem of the excited tuning fork (forced vibrations).
c) Vibration Frequency: The table vibrates at the frequency of the external driving force, which is 288 Hz.

Question 12. B. (OR)

When an excited tuning fork of natural frequency 256 Hz has its stem pressed on a metal plate of the same frequency, the plate vibrates.
a) What will be the frequency of vibration of the plate?
b) What would be the change in the vibration if the natural frequency of the plate is 200 Hz? Justify your answer.

Answers:
a) Frequency of Plate: 256 Hz
b) Change in Vibration: The amplitude of vibration of the plate will decrease significantly, and the sound produced will become much fainter.

> ***Justification:*** *Initially, the frequencies matched, creating resonance (maximum amplitude). At 200 Hz, resonance no longer occurs, leaving only standard forced vibrations with small amplitude.*

Question 13

Write down three differences between transverse waves and longitudinal waves.

Feature	Transverse Waves	Longitudinal Waves
Particle Vibration	Particles vibrate perpendicular to the direction of wave propagation.	Particles vibrate parallel to the direction of wave propagation.
Structure	Formed of alternating crests and troughs.	Formed of alternating compressions and rarefactions.
Medium	Can travel through solids and on liquid surfaces.	Can travel through all states of matter (solids, liquids, and gases).

Question 14

Bats use ultrasonic sound for navigation.
a) Write down four instances where we use ultrasonic waves.
b) Explain what ultrasonography is.

Answers:
a) Four Instances of Ultrasound Use:
1. SONAR: To map ocean floors and detect underwater submarines.
2. Medical Diagnostics: For ultrasonography to monitor fetal growth.
3. Industrial Cleaning: To clear dust/grease from intricate mechanical parts.
4. Flaw Detection: To scan for internal cracks inside solid metal blocks.
b) Ultrasonography: A medical imaging technique that sends high-frequency ultrasound waves into the body to create real-time internal images of organs based on reflected echoes.

Question 15. A.

Can human hear a wave produced by the oscillation of an object of period 1 s? Justify your answer.

Answer: No, humans cannot hear this sound.

> ***Justification:*** *Frequency (f) = 1 / Time Period (T) = 1 / 1 s = 1 Hz.*
> *The audible range for normal human hearing is 20 Hz to 20,000 Hz. Since 1 Hz falls below 20 Hz (Infrasonic region), it is completely inaudible.*

Question 15. B. (OR)

The wavelength of a sound wave travelling with a speed of 350 m/s is 1 cm. How many regions of high pressure will this sound produce in one second? Can we hear this sound? Justify your answer.

Answer / Calculation:

- **Speed (v) = 350 m/s**
- **Wavelength (λ) = 1 cm = 0.01 m**
- **Frequency (f) = v / λ = 350 / 0.01 = 35,000 Hz**

Part 1: Frequency represents cycles per second. Since each cycle has one compression, it will produce 35,000 high-pressure regions in one second.

Part 2: No, we cannot hear this sound.

Justification: 35,000 Hz is greater than the upper human limit of 20,000 Hz, making it an inaudible ultrasonic sound.